TEXORA · LECTURE SHEET

Sweater Manufacturing — Fully Fashioned Article Manufacturing

The sweater knitted to shape, not cut to shape: V-bed flat machines, shaping by stitch transfer, the panel set, and the linking, finishing and pressing line that turns shaped panels into a garment with fashioning marks as its quality signature.

FM-4201 · ADVANCED KNITTING AND NON-WOVEN · TOPIC 2 · FABRIC MANUFACTURING SPECIALIZATION

1. Three ways to make a sweater

ROUTE	HOW PARTS ARE OBTAINED	WASTE	QUALITY SIGNATURE
Cut-and-sew	Rectangular fabric knitted, garment pieces cut out	Highest (cutting loss)	Seams cut through loops; cheapest; volume basics
Fully fashioned	Each panel (front, back, sleeves, collar) knitted to shape by increasing/decreasing	Low (selvedge trim only)	**Fashioning marks** at shaped edges; seams join loop to loop
Whole garment	Complete garment knitted connected (Topic 1)	Lowest	Near-seamless; highest machine/programme demand

This topic treats the middle route — the classic quality sweater industry of flat-bed machines — which also supplies every concept the whole-garment route automates.

2. The machine and its gauge

The **V-bed flat knitting machine** carries two needle beds inclined in an inverted V, with latch needles sliding in tricks, worked by a traversing carriage whose cam systems knit, tuck, miss, transfer and (on modern machines) split stitches. **Gauge** is needles per inch: fine gauges (12–14 gg and finer) for lightweight sweaters, mid (7–10 gg) for classic knitwear, coarse (3–5 gg) for chunky products. Gauge fixes the stitch scale of everything: at 12 gg a panel needing 45 cm of width needs $45 \times (12 \div 2.54) \approx$ **213 stitches** — the number the programme starts from (worked below). Presser feet/sinker systems hold the fabric down; takedown rollers draw the knitted panel away at controlled tension, which is what makes reliable shaping possible.

3. Shaping by transfer: the fashioning principle

A panel is shaped at its edges by **decreasing** (transferring edge stitches inward so two stitches share a needle — narrowing) and **increasing** (bringing empty needles into work — widening). Decreasing done one or more stitches in from the edge leaves a visible decorative line of overlapped stitches — the **fashioning mark**, the hallmark that a panel was shaped, not cut. Armholes, sleeve tapers, neck openings and shoulder slopes are all written as schedules of shaping events against course counts.

WORKED EXAMPLE — A SLEEVE NARROWING SCHEDULE

A sleeve must lose 20 stitches *at each edge* over a shaping zone of 100 courses, decreasing 1 stitch per edge at each shaping event.

Decreases per edge = 20 → shaping events = 20; spacing = 100 ÷ 20 = **one decrease each side every 5 courses**.

The programmer states it exactly so — "dec 1 st each side every 5th course, 20 times" — and the same arithmetic, sign reversed, writes widening zones. Every fashioned measurement in the garment spec (lengths, widths at stated courses) reduces to these two numbers: how many stitches, over how many courses.

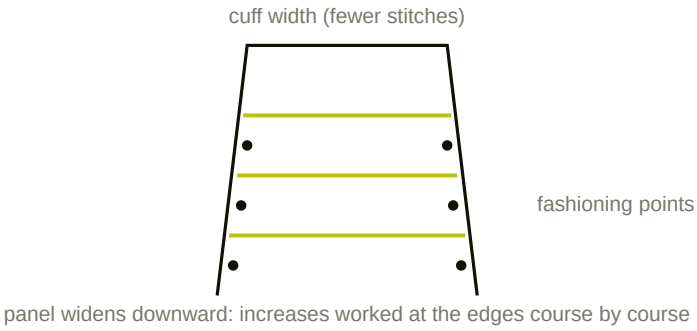


Figure 1 — A fashioned sleeve panel in outline (schematic): the tapered edges are created stitch by stitch; the dots mark shaping events whose spacing (worked example: every 5 courses) is calculated, not eyeballed.

4. The panel set and the make-up line

A sweater order produces a **panel set**: front(s), back, two sleeves, collar/neck band, and trims (pockets, button bands as applicable), each knitted to its own programme, weighed and checked against the spec (panel weight is the fastest whole-panel quality check — it integrates count, stitch length and dimensions). The make-up sequence then runs:

1. **Linking (looping):** panels joined stitch-to-stitch on a linking machine whose operator aligns corresponding loops on the point ring — the seam that keeps fashioned knitwear's elasticity and flat profile; the skilled bottleneck of the classic factory.

2. **Seaming of remaining joins** (overlock/cup seaming where linking is not specified), cuff and hem finishing, buttonholing and buttoning.
3. **Washing/milling:** relaxes the panels, removes knitting oils, evens the loops and, for wool, can add a controlled milled (lightly felted) finish; dimensions are finally set here — panel specs are written for the *finished* garment, so knitting targets anticipate wash shrinkage (the dimensional behaviour studied practically in FM-4202 Topic 5).
4. **Boarding/pressing:** the garment is drawn onto heated forms (boarding) or steam-pressed to the measurement table.
5. **Inspection and measurement:** against the customer's measurement chart (chest, body length, sleeve length, neck), shade banding between panels of one garment, and fault grading; then folding and packing.

Quality pressure points follow the physics: yarn lots change → panel weight drifts; stitch length drifts → dimensions drift; transfer faults → dropped stitches at fashioning points. The fully fashioned factory is therefore run on measurement: panel weight per programme revision, course counts, and finished-garment sampling — a smaller, more skilled operation than cut-and-sew, selling the difference as fit, seam quality and longevity.

Glossary

Fully fashioned — panels knitted to shape by increasing/decreasing. **Fashioning mark** — the decorative stitch line left by in-from-edge decreases. **Gauge (gg)** — needles per inch of a flat machine. **Linking** — stitch-to-stitch joining of panels on a point ring. **Boarding** — setting garment shape on heated forms. **Panel set** — all knitted components of one garment.

Key points

• Fully fashioned = shape by transfer: every measurement becomes "how many stitches over how many courses" (worked: 1 st each side every 5 courses × 20).
• Gauge sets stitch scale: 12 gg → ≈213 stitches for 45 cm width.
• The make-up line — linking, washing, boarding, measurement — is where fashioned quality is kept or lost; panel weight is its master check.
• Against cut-and-sew it trades machine/programme/skill cost for low waste, elastic loop-to-loop seams and fit.

Practice

1. A front panel must widen by 12 stitches at each side over 48 courses, increasing 1 stitch per side per event. Write the shaping instruction.
2. How many stitches are needed for a 50 cm panel width on a 7 gg machine?
3. Why is panel weight such a powerful routine check in a fashioned-knitwear factory?
4. State the order of operations from knitted panel to packed garment, and name the stage at which final garment dimensions are set.

Answer key

1. Increases per side = 12 → events = 12; spacing = 48 ÷ 12 = 4 → **"inc 1 st each side every 4th course, 12 times"**.
2. Stitches = 50 × (7 ÷ 2.54) = 50 × 2.76 ≈ **138 stitches**.
3. Panel weight integrates everything that can drift — yarn count, stitch length (via yarn consumed per course), course count and width — into one number measurable in seconds at the machine. A weight outside tolerance announces a fault before anyone measures a dimension, and it does so per panel, per machine, per shift.
4. Linking/joining → subsidiary seaming and trims → washing/milling → boarding/pressing → inspection/measurement → folding/packing. Final dimensions are set in **washing and boarding**: knitting is targeted at pre-wash dimensions calculated to land on the measurement table after relaxation shrinkage.
